Apply: Industry AI Challenge, From Pre-trained Model to Business Solution

Start Assignment

- Due Wednesday by 11:59pm
- Points 20
- Submitting a file upload
- Available Nov 2 at 12am - Nov 11 at 11:59pm

Imagine your team has been hired as an AI consulting team by a company that wants to solve a real operational problem using artificial intelligence. The company is not interested in AI simply because it is interesting. It wants a solution that can improve an existing process, reduce manual work, save money, improve customer experience, or support better decisions. Your responsibility is to investigate whether a pre-trained deep learning model can provide a practical solution. You will select a real-world problem, identify a suitable pre-trained model, adapt it to your dataset, conduct controlled experiments, and recommend a solution based on your findings. **Your goal is not necessarily to build the most accurate model. Your goal is to make and defend a sound engineering recommendation.** I will evaluate your reasoning, experimentation, technical understanding, collaboration, and ability to explain your findings to a nontechnical stakeholders. I have assigned the following teams:

Team	Members
1	Mataeo Anderson and Jesse Mazariegos
2	Jeffrey Gregory and Jake Van Slyke
3	Jolie Walker and Cooper Gunnels

I am not assigning individual technical roles. Your team may coordinate responsibilities, but no part of the project should become the exclusive responsibility or knowledge of one person. Every member must participate in major technical decisions and understand the complete project. I may ask any team member about the problem, dataset, model, experiments, results, and final recommendation.

1. Understand the Business Problem

Before selecting a model, identify a business problem worth solving. Your problem must have a clear user, stakeholder, or organization that could benefit from your solution.

Examples include:

Industry	Potential problem	Business value
Manufacturing	Detect defective products from images	Reduce manual inspection

Retail	Classify product images into categories	Improve inventory management
Customer service	Automatically categorize customer support tickets	Reduce response delays
Transportation	Identify vehicle damage from photographs	Support inspection workflows
E-commerce	Categorize customer reviews or complaints	Identify recurring issues
Agriculture	Identify crop diseases using images	Support early detection

You are welcome to choose a different problem.

Required: Business Problem Brief

In approximately half a page, explain:

1. Who is your hypothetical client or stakeholder?
2. What problem are they trying to solve?
3. How is the problem currently handled, or how might it be handled without AI?
4. What would make your AI solution useful to the organization?
5. What would happen if the model made an incorrect prediction?

Important: You do not need to contact a real company. You may construct a realistic business scenario, but your technical investigation must use a real, publicly available dataset.

2. Select a Model and Dataset

Select a publicly available pre-trained deep learning model suitable for your problem.

Possible sources include:

- Keras Applications
- TensorFlow Hub
- Hugging Face
- Kaggle Models
- Other credible model repositories

Select a documented public dataset that is legal to use.

Your proposal must explain:

- Which pre-trained model you selected and why.
- What another reasonable model option would be.
- What your dataset contains, including its size and target structure.
- What limitations the dataset has relative to the intended business application.
- How you will adapt the pre-trained model to your problem.

You may use transfer learning, feature extraction, partial fine-tuning, or full fine-tuning when computationally feasible. Simply downloading a model and running predictions is not sufficient.

Computational constraint: Select a model and dataset that allow you to conduct several experiments. If training takes more than approximately 30 minutes per experiment, consider reducing the dataset or using a smaller model.

3. Define Success Before Training

Industry teams do not evaluate models using accuracy alone. Their performance requirements depend on how the model will be used. Before running experiments, identify:

A. Model success metric

Choose an appropriate primary metric such as precision, recall, F1-score, accuracy, MAE, or IoU. Explain why the metric matters for your problem.

B. Business success criterion

Define one realistic target that would make the model useful. For example:

- A defect detection system should miss as few defective products as reasonably possible.
- A customer support classifier should correctly identify urgent complaints.
- An inventory classification system should minimize unnecessary manual review.

C. Operational constraint

Identify at least one practical constraint, such as inference time, model size, compute requirements, or the ability to run on commonly available hardware. You do not need to prove that your solution is production-ready. You do need to explain what would matter if the organization wanted to use it.

4. Build a Baseline and Conduct Controlled Experiments

You must complete at least four runs: **One baseline + three controlled experiments.**

Your first run should establish a working reference configuration using your selected pre-trained model. You must also include an appropriate simple benchmark, such as majority-class prediction for classification. For the three experiments, change one major variable at a time.

Examples include:

- Frozen versus partially trainable layers.
- Different learning rates.
- Different preprocessing strategies.
- Different input resolutions.
- Different augmentation techniques.
- Different classification heads.
- Different batch sizes.
- Alternative optimization settings.

Use the same train/validation/test split across experiments when possible. Make model-selection decisions using validation data and reserve the test set for final evaluation.

Experiment Log

Maintain the following log as you work:

Run	Change made	Reason for change	Model result	Practical trade-off
Baseline	Initial configuration	Establish reference		
Experiment 1				
Experiment 2				
Experiment 3				

For each experiment, explain what you expected, what happened, and what you learned. I am more interested in your ability to interpret evidence than in whether every experiment improves performance.

5. Evaluate the Solution Like an ML Engineer

Your evaluation must address three questions.

A. How well does the model perform?

Include appropriate evidence such as:

- Training and validation curves.
- Relevant evaluation metrics.
- Confusion matrix, if appropriate.
- Correct and incorrect predictions.
- Comparison of experimental results.

B. Where might the model fail?

- Identify and analyze at least three incorrect or difficult predictions, if available.

- Discuss possible reasons for these errors and their consequences for the organization.
- If your model misses a defective product, for example, is that more serious than incorrectly flagging a good product?
- Explain how such considerations influence your recommended model.

6. Analyze an Unsuccessful Experiment

Describe at least one genuine experiment or development attempt that produced a worse-than-expected result.

Explain:

1. Why you believed the change might improve the model.
2. What happened.
3. What evidence helps explain the result.
4. What your team learned.
5. Whether the result changed your final recommendation.

Do not intentionally create a failing configuration simply to complete this requirement. Unsuccessful experiments are a normal part of professional ML development.

7. Make an Industry Recommendation

Your team must make a final recommendation to its hypothetical client. Please do not simply select the configuration with the highest accuracy. You need to consider model performance, stability, false predictions, inference efficiency, and the business requirements you identified. Your recommendation must answer:

Would you recommend this AI solution for a limited pilot? Why or why not?

You may recommend one of three outcomes:

- **Proceed with a limited pilot:** The results are sufficiently promising for further real-world evaluation.
- **Continue development:** The approach has potential but needs additional improvement.
- **Do not proceed:** The present approach does not adequately address the problem.

8. Team Collaboration and Accountability

I have already assigned the project teams. Your team may coordinate responsibilities, but no major technical component should become the exclusive responsibility or knowledge of one member. Every member should be responsibile for the whole project.

Team Charter — Half Page

Submit one signed charter per team covering by Nov 5, 2026:

- How and when your team will collaborate.
- Where your code and documents will be shared.
- How technical decisions will be made.
- How you will ensure shared understanding.
- How you will handle blockers or uneven contributions.

Individual Contribution Statement — Half Page

Each student must independently explain:

1. What they contributed to the project.
2. One technical decision they helped make.
3. Something they learned from a teammate.
4. One challenge the team encountered and how they helped resolve it.
5. What they would improve if the project continued.

Peer Evaluation

Each student will privately evaluate themselves and their teammate(s) on a 1–4 scale in three areas:

- Active and reliable contribution.
- Effective communication.
- Initiative and leadership.

Provide a brief justification for each rating. I may ask any team member questions about any part of the project.

9. Final Deliverables

#	Deliverable	Format
1	Industry project report	PDF, 4–5 pages including key figures; references may be additional
2	Executed notebook or source code	`.ipynb` or accessible repository
3	Experiment comparison and evidence	Included in report/notebook
4	Business recommendation	Included in report

5	Team charter	Half page, signed and submitted separately.
6	Individual contribution statement	Half page, individual
7	Peer evaluation	Individual, submitted by each student
8	Industry pitch	6–8 minutes per team

Industry Pitch

Please treat your final presentation as a meeting with a potential client. Your audience includes people who understand the business problem but may not understand deep learning.

Your presentation should explain:

1. The business problem and why it matters.
2. Why your team selected the model.
3. What experiments you conducted.
4. What your results show.
5. The model's important limitations and practical trade-offs.
6. Your final recommendation.

Every team member must participate and be prepared to answer technical questions.

10. Assessment Criteria

Component	Weight
Business problem framing and success criteria	15%
Technical approach and model selection	15%
Controlled experimentation and reproducibility	25%
Results, error analysis, and practical constraints	20%
Final recommendation and industry pitch	15%
Collaboration and individual understanding	10%
Total	**100%**